5.3.2 Definite & Indefinite Integrals

Motivation: In the last section we saw that antiderivatives are important for computing definite integrals. We would like some formal notation for the most general antiderivative of a function.

Definition: We use an **indefinite integral** of *f*, written as $\int f(x)\,dx$, to indicate the most general antiderivative of *f*.

Example 1: Find each of the following.

a) Find $\int x^2 dx$.

We can easily see that $\int x^2 dx = \frac{1}{3}x^3 + C$

b) Find $\int_0^3 x^2 dx$

We note that $f(x) = x^2$ is a continuous function on the interval $[0,3]$. Therefore by the FTCP2 we know that:

$$\int_0^3 x^2 dx = \frac{1}{3}x^3\Big|_0^3 = \frac{1}{3}(3)^3 - \frac{1}{3}(0)^3 = 9 - 0 = 9$$

Note: VERY IMPORTANTLY, we see that an indefinite integral represents a family of functions, while a definite integral represents an area

Properties of Indefinite Integrals (Antiderivatives)

(1) $\int c \cdot f(x)\,dx = c \cdot \int f(x)\,dx$ (constant multiple rule in reverse)

(2) $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$ (sum & difference rule in reverse)

(3) $\int f(x)g(x)\,dx \neq \int f(x)\,dx \cdot \int g(x)\,dx$ (you *cannot* take the antiderivative of each piece of a product) This is *NOT* a derivative rule in reverse.

In the online textbook there is a link to what is called the "Table of Integrals" and it is very important to know ALL of these formulas for Calculus I, II, & III. All of these formulas simply represent "differentiation in reverse".

General Strategies for Finding Antiderivatives

(1) Use a derivative rule in reverse.

(2) Use Algebra to simplify your expression and then use (1).

Note: We will see here that "Algebra" includes things such as simplifying expressions, factoring, using trig. identities, and a variety of other simplification techniques.

Note: The list above is by no means complete. We will continue to add to this list throughout the remainder of Calculus I and during Calculus II.

Example 2: Evaluate $\int \frac{1}{\sqrt{1-x^2}}\,dx$.

This is simply a derivative rule in reverse and so this question is easy to answer: $\int \frac{1}{\sqrt{1-x^2}}\,dx = \sin^{-1}(x) + C$

Note: In Examples 3 – 6 (below) a variety of different "Algebra" techniques are shown.

Example 3: Evaluate $\int \frac{x^3 - 16x^4\sec^2(x) + 8}{8x^4}\,dx$. ***(Split Fractions)***

$$\begin{aligned}
\int \frac{x^3 - 16x^4\sec^2(x) + 8}{8x^4}\,dx &= \int\left[\frac{x^3}{8x^4} - \frac{16x^4\sec^2(x)}{8x^4} + \frac{8}{8x^4}\right]dx = \int\left[\frac{1}{8x} - 2\sec^2(x) + \frac{8}{8x^4}\right]dx \\
&= \int\left[\frac{1}{8}\cdot\frac{1}{x} - 2\sec^2(x) + x^{-4}\right]dx \\
&= \int \frac{1}{8}\cdot\frac{1}{x}\,dx - \int 2\sec^2(x)\,dx + \int x^{-4}dx \\
&= \frac{1}{8}\int \frac{1}{x}\,dx - 2\int \sec(x)\,dx + \int x^{-4}dx \\
&= \left(\frac{1}{8}\ln|x| + C_1\right) - \left(2\tan(x) + C_2\right) - \left(\frac{1}{3}x^{-3} + C_3\right) \\
&= \frac{1}{8}\ln|x| - 2\tan(x) - \frac{1}{3}x^{-3} + C
\end{aligned}$$

Note: When working through most problems we will skip some of the detail we used in the example above. Often we will skip from line 3 to the solution in a single step when possible. It is expected that you should understand and be able to visualize the intermediate steps even though we will not perform them.

Example 4: Evaluate $\int_1^2 \frac{x^2+4x+3}{x+1}\,dx$. ***(Factoring)***

This is a definite integral so we are looking for an area.

Since the function $f(x)=\frac{x^2+4x+3}{x+1}$ is continuous on $[1,2]$, so by the FTCP2 we know that:

$$\int_1^2 \frac{x^2+4x+3}{x+1}\,dx = \int_1^2 \frac{(x+3)(x+1)}{x+1}\,dx = \int_1^2 (x+3)\,dx = \left(\frac{1}{2}x^2+3x\right)\Bigg|_1^2$$

$$=\left[\frac{1}{2}(2)^2+3(2)\right]-\left[\frac{1}{2}(1)^2+3(1)\right]$$

$$=[2+6]-\left[\frac{1}{2}+3\right]$$

$$=4.5$$

Example 5: Evaluate $\int \frac{x^2+10}{x^2+1}\,dx$. ***(Split fractions cleverly)***

$$\int \frac{x^2+10}{x^2+1}\,dx = \int \frac{x^2+1+9}{x^2+1}\,dx = \int\left[\frac{x^2+1}{x^2+1}+\frac{9}{x^2+1}\right]dx = \int\left[1+\frac{9}{x^2+1}\right]dx = x+9\tan^{-1}(x)+C$$

Example 6: Evaluate $\int_{-1}^0 \frac{x^2}{x^2+1}\,dx$. ***(Introduce a "fancy 0")***

This is a definite integral so we are looking for an area.

Since the function $f(x)=\frac{x^2}{x^2+1}$ is continuous on $[0,-1]$, so by the FTCP2 we know that:

$$\int_{-1}^0 \frac{x^2}{x^2+1}\,dx = \int_{-1}^0 \frac{x^2+1-1}{x^2+1}\,dx = \int_{-1}^0\left[\frac{x^2+1}{x^2+1}-\frac{1}{x^2+1}\right]dx = \int_{-1}^0\left[1-\frac{1}{x^2+1}\right]dx = \left[x-\tan^{-1}(x)\right]\Big|_{-1}^0$$

$$=\left[0-\tan^{-1}(0)\right]-\left[1-\tan^{-1}(-1)\right]=[0-0]-\left[1-\left(-\frac{\pi}{4}\right)\right]$$

$$=1-\frac{\pi}{4}$$

Important Trig. Identities to Know:

- Pythagorean Identities
  - $\sin^2(x)+\cos^2(x)=1$
  - $\tan^2(x)+1=\sec^2(x)$
  - $1+\cot^2(x)=\csc^2(x)$

- Double/Half Angle Identities
  - $\sin(2x)=2\sin(x)\cos(x)$
  - $\cos(2x)=\cos^2(x)-\sin^2(x)$

Example 7: Evaluate $\int \sec(x)\left[\sec(x)+\tan(x)\right]dx$.

$$\int \sec(x)\left[\sec(x)+\tan(x)\right]dx=\int\left[\sec^2(x)+\sec(x)\tan(x)\right]dx=\tan(x)+\sec(x)+C$$

Example 8: Evaluate $\int \tan^2(x)\,dx$.

$$\int \tan^2(x)\,dx=\int\left[\sec^2(x)-1\right]dx=\tan(x)-x+C$$

Example 9: Evaluate $\displaystyle\int_{\pi/4}^{\pi/2}\frac{\cos(\theta)}{\sin^2(\theta)}d\theta$.

This is a definite integral so we are looking for an area.

Since the function $f(\theta)=\dfrac{\cos(\theta)}{\sin^2(\theta)}$ is continuous on $\left[\dfrac{\pi}{4},\dfrac{\pi}{2}\right]$, so by the FTCP2 we know that:

$$\int_{\pi/4}^{\pi/2}\frac{\cos(\theta)}{\sin^2(\theta)}d\theta=\int_{\pi/4}^{\pi/2}\frac{1}{\sin(\theta)}\cdot\frac{\cos(\theta)}{\sin(\theta)}\cdot d\theta=\int_{\pi/4}^{\pi/2}\csc(\theta)\cot(\theta)\,d\theta=-\csc(\theta)\Big|_{\pi/4}^{\pi/2}=-\csc\left(\frac{\pi}{2}\right)+\csc\left(\frac{\pi}{4}\right)=-1+\sqrt{2}$$